

AI Readiness Scale (AIRS)

Development and Validation of a Psychometric Instrument for Assessing Enterprise AI Adoption

Doctor of Business Administration Dissertation Defense

DBA DISSERTATION

N = 523 · OCTOBER–NOVEMBER 2025



Agenda

Dissertation Overview

01

Research Problem & Significance

The adoption-value gap and three motivating gaps

02

Theoretical Framework

UTAUT2 extended with AI Trust

03

Methodology

Mixed-methods design, instrument development, and analytical pipeline

04

Key Findings

Psychometric properties, structural model, moderation, and typology

05

Contributions & Implications

Theory, practice, and future research roadmap

The Adoption-Value Gap

88%

AI Tool Usage

Knowledge workers using
AI tools in 2025, up from
50% in 2023

5%

Value Realized

Organizations reporting
measurable value from AI
implementations

90-...

Pilots Stalled

Generative AI pilot
programs that do not
progress beyond initial
trials

Organizations invest heavily in AI tools yet consistently fail to realize meaningful returns – a critical problem for both scholarship and practice.



Three Motivating Gaps

Theoretical Gap

UTAUT2 was validated for consumer mobile technologies. Whether its predictor structure holds for AI – characterized by autonomous decision-making, probabilistic outputs, and evolving capabilities – remained empirically untested.

Measurement Gap

No validated psychometric instrument existed for AI-specific adoption readiness. Existing scales addressed general technology acceptance or narrow AI applications, leaving organizations without a diagnostic tool.

Practice Gap

Organizations lacked evidence-based frameworks to identify specific adoption barriers and design targeted interventions. Without systematic measurement, AI implementation strategies remained generic rather than responsive.

Chapter 1

Theoretical Framework

The study builds upon UTAUT2 – the most comprehensive validated model of technology acceptance – and extends it with an AI-specific construct to address the unique characteristics of artificial intelligence as a technology category.



UTAUT2 Foundation: Seven Constructs

Performance Expectancy (PE)

Perceived improvement in job performance from using the technology

Effort Expectancy (EE)

Perceived ease of learning and using the technology

Social Influence (SI)

Degree to which important others believe one should use the technology

Facilitating Conditions (FC)

Perception of available resources and support infrastructure

Hedonic Motivation (HM)

Pleasure or enjoyment derived from using the technology

Price Value (PV)

Perceived balance between benefits received and monetary cost

Habit (HB)

Extent to which technology use has become automatic

AI-Specific Extension: AI Trust

AI tools differ fundamentally from conventional technologies in their capacity for **autonomous reasoning**, opaque decision processes, and evolving behaviors.

AI Trust (TR) was introduced as an eighth construct, capturing the degree to which users believe AI systems will provide accurate, reliable information and operate in a trustworthy manner.

This extension is grounded in emerging literature identifying trust as a unique barrier to AI adoption not adequately captured by existing UTAUT2 constructs.



📄 **Key Distinction:** Trust in AI is not merely confidence in a tool — it reflects users' beliefs about autonomous, probabilistic systems that may behave unexpectedly.

Research Questions & Hypotheses

1

Primary RQ

To what extent does UTAUT2, extended with AI Trust, explain Behavioral Intention to adopt AI tools among U.S. students and professionals?

2

RQ2: Factorial Validity

Psychometric adequacy of the extended model

3

RQ3: Moderation

Experience and role moderation of structural paths

4

RQ4 & RQ5

Behavioral intention–usage relationship and user typology based on readiness patterns

Twelve hypotheses were tested: seven core UTAUT2 path hypotheses (H1a–H1g), one AI Trust hypothesis (H2), two moderation hypotheses (H3–H4), and two behavioral validation hypotheses (H5–H6).

Chapter 2

Methodology

Design

Post-positivist sequential mixed-methods with quantitative primacy

Sample

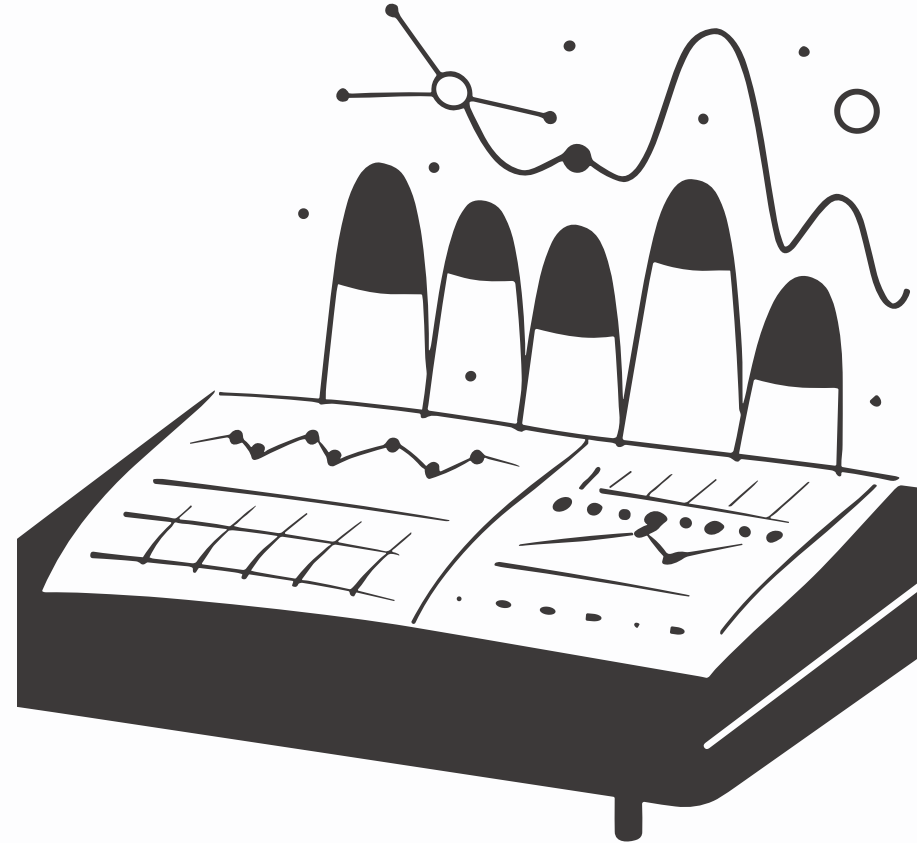
N = 523 U.S. adults via Centiment topic-blinded recruitment

Period

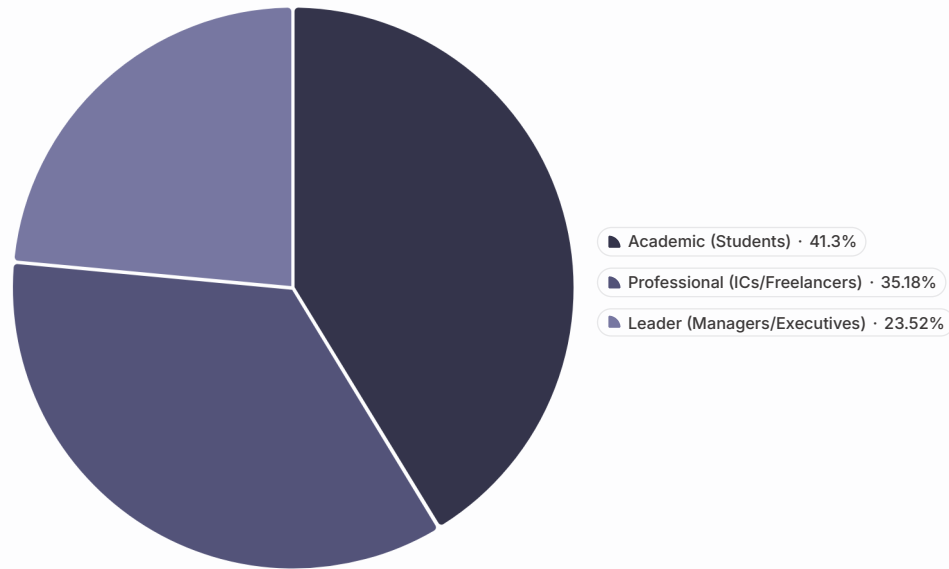
October–November 2025 (3-week collection window)

Platform

Dual-platform validation: Python + R/lavaan 0.6.21



Sample Composition



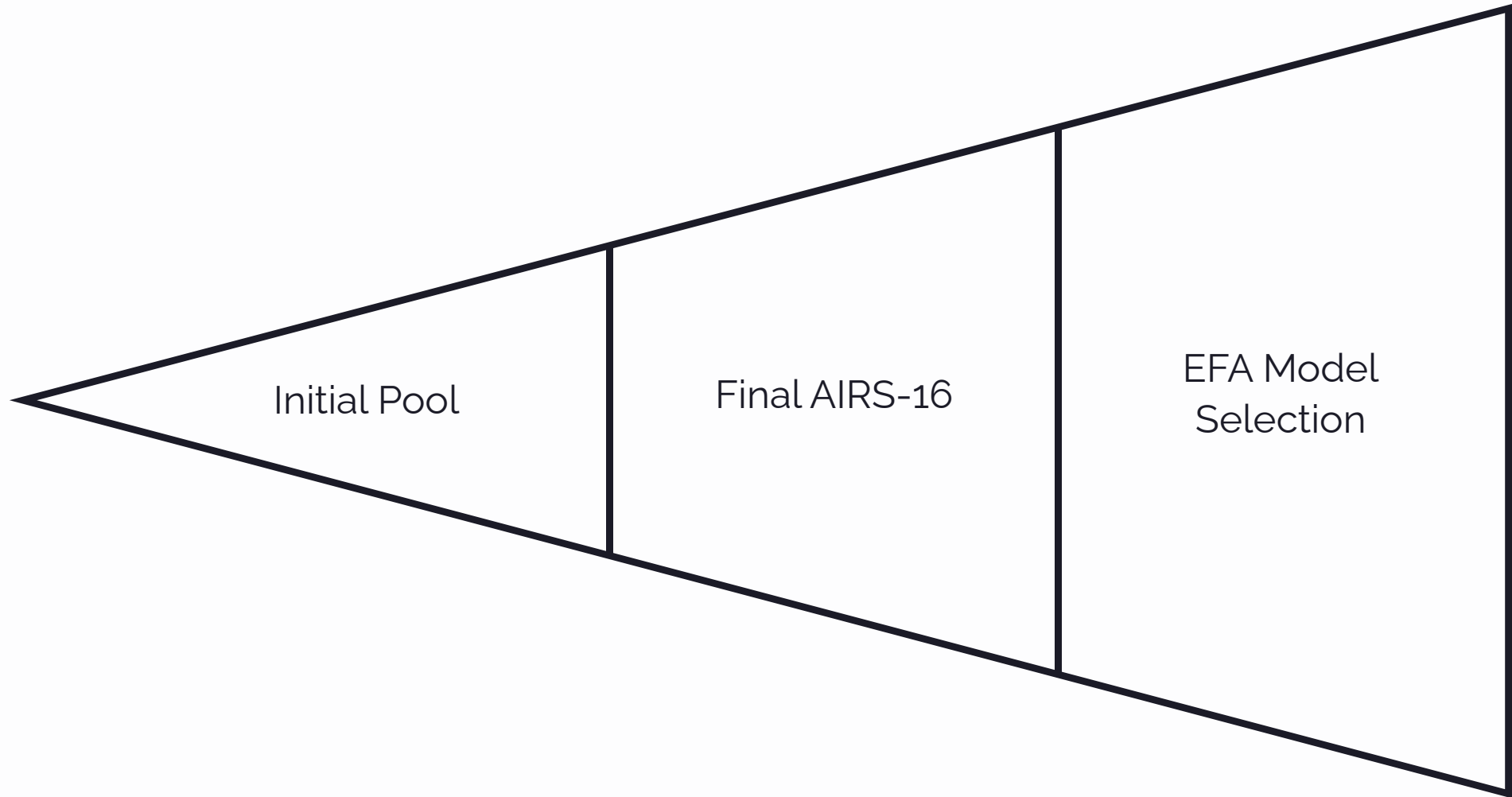
Key Sampling Decisions

Topic-blinded recruitment via Centiment: respondents did not know the survey concerned AI when deciding to participate, mitigating self-selection bias.

50/50 sample split (seed = 67, stratified by AI adoption): development subsample (n = 261) for EFA; holdout sample (n = 262) for CFA confirmation.

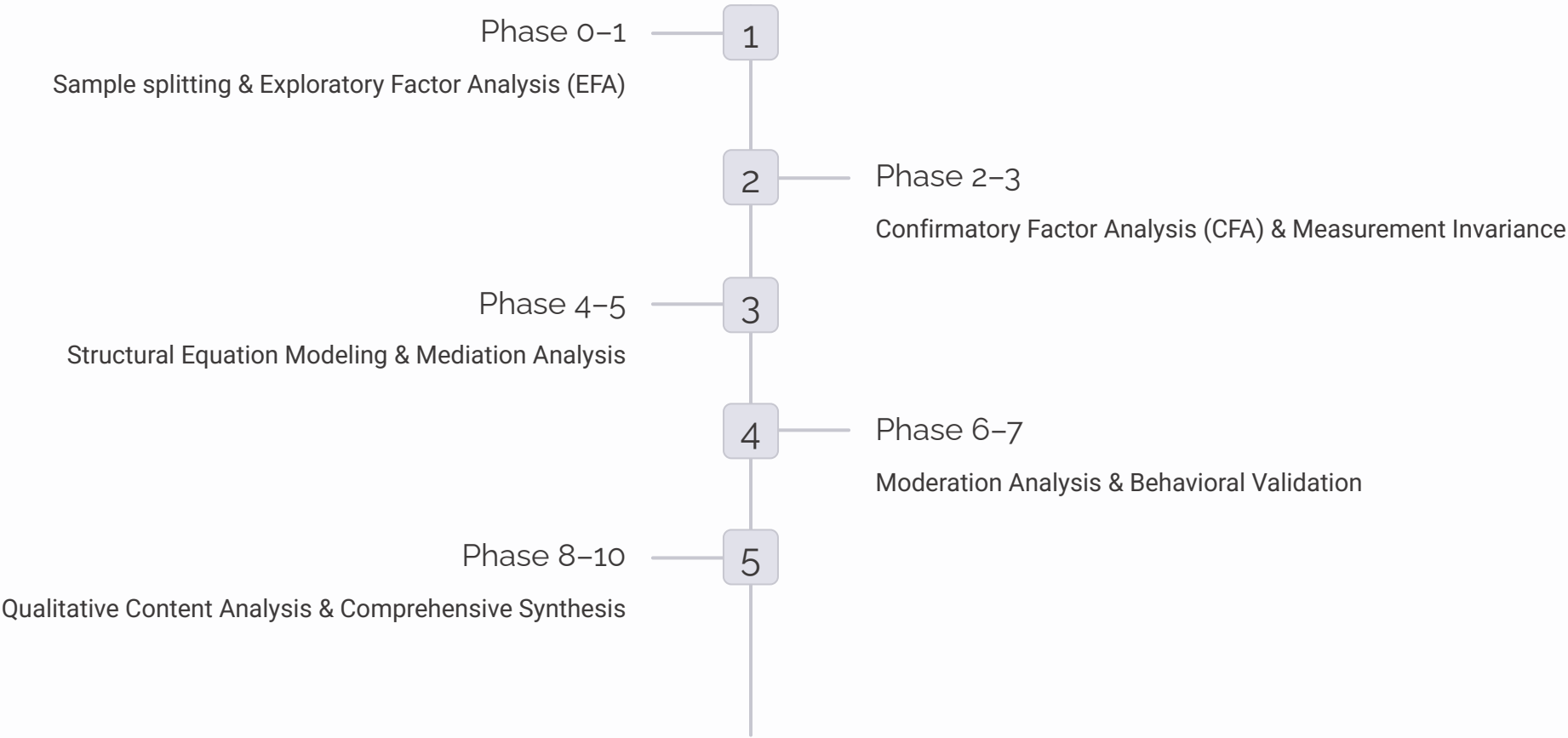
Disability representation: 68 participants (13.0%) self-reported a disability — consistent with U.S. Census Bureau estimates.

Instrument Development: AIRS

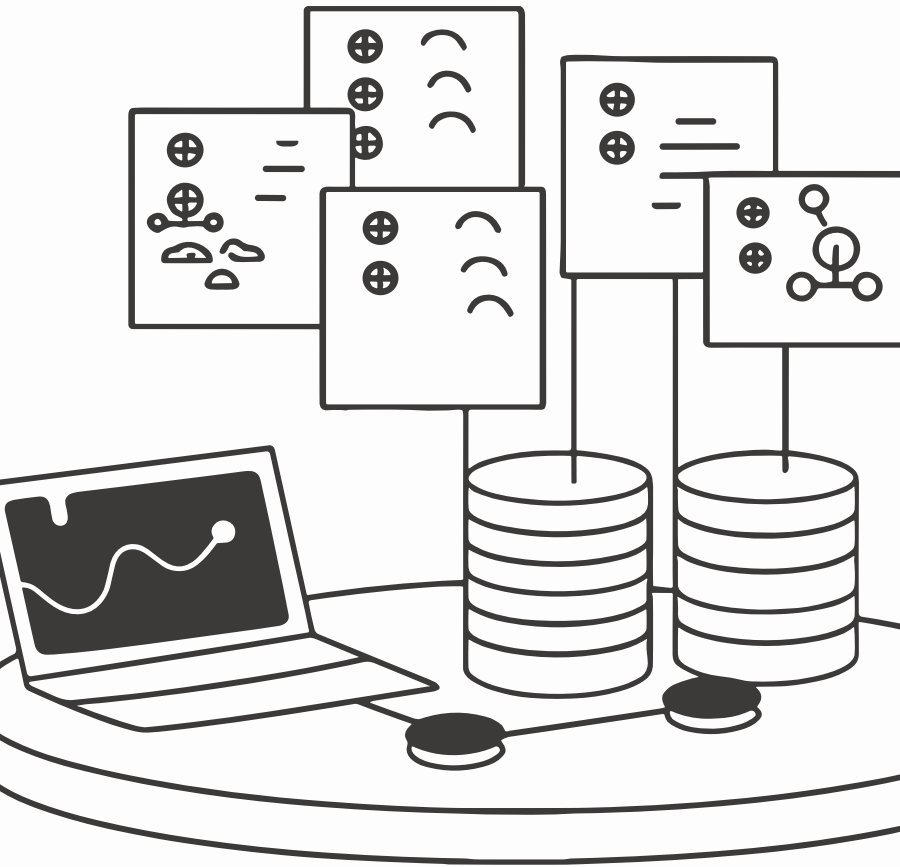


The final instrument retains 8 theoretically interpretable factors with strong psychometric properties, providing a compact yet diagnostically rich tool for organizational assessment.

10-Phase Analytical Pipeline



EFA used MINRES extraction with promax rotation (item retention: primary loading $\geq .50$, cross-loading gap $\geq .20$, communality $\geq .30$). CFA/SEM used Maximum Likelihood with Satorra-Bentler robust corrections (MLM). Bootstrap validation: 1,000 resamples.



Chapter 3

Key Findings: Measurement Model

The 16-item AIRS demonstrated excellent suitability for factor analysis: **KMO = .937** (superb) and Bartlett's Test $\chi^2 = 4,668.45$, $p < .001$. No multivariate outliers were detected.

CFA Model Fit: Holdout Sample (n = 262)

Index	Value	Threshold	Assessment
CFI	.975	≥ .95	Excellent
TLI	.960	≥ .95	Excellent
RMSEA	.065	≤ .08	Good
SRMR	.026	≤ .08	Excellent
χ^2/df	2.10	≤ 3.0	Good

❏ Full-sample CFA confirmed: CFI = .979, TLI = .966, RMSEA = .061, SRMR = .022

What This Means

The 8-factor model achieved **excellent fit** on an independent holdout sample — a stringent test that rules out overfitting to the development data.

Cross-validation across dual platforms (Python and R/lavaan) confirmed the stability of these results, providing high confidence in the instrument's psychometric integrity.

Reliability & Convergent Validity

Factor	Cronbach's α	Composite Reliability	AVE	Status
Performance Expectancy	.863	.866	.764	✓
Effort Expectancy	.803	.805	.674	✓
Social Influence	.792	.793	.659	✓
Facilitating Conditions	.743	.750	.601	✓
Hedonic Motivation	.909	.910	.835	✓
Price Value	.882	.883	.791	✓
Habit	.909	.909	.833	✓
AI Trust	.862	.862	.758	✓

All composite reliabilities exceeded .750 and all AVE values exceeded .50, confirming convergent validity across all eight factors.

Discriminant Validity: HTMT Analysis

HTMT Violations (Threshold: .85)

Pair	HTMT	Status
PE × PV	.902	Exceeds
PE × HM	.900	Exceeds
HM × PV	.904	Exceeds
HM × TR	.850	At threshold

Interpretive Context

Violations are concentrated in the **PE–HM–PV triad** and are structurally expected for 2-item-per-factor scales with conceptually adjacent constructs.

The CFA model nonetheless demonstrated excellent overall fit. Each construct measures a **distinct intervention target** — organizations benefit from separately assessing performance perceptions, enjoyment, and value even when these dimensions are highly correlated.

❏ HTMT violations do not compromise the instrument's diagnostic purpose.

Chapter 4

Structural Model Results

89.7%

Variance Explained
 R^2 for Behavioral Intention
— an exceptionally high value indicating AIRS captures the vast majority of systematic variance in AI adoption intention

12

Hypotheses Tested
Across UTAUT2 core paths, AI Trust extension, moderation, and behavioral validation

5

Supported
Hypotheses fully or marginally supported, with 3 partially supported and 4 not supported



Primary Path Coefficients

H	Path	β	p	Bootstrap 95% CI	Result
H1a	PE → BI	-.028	.791	[-.234, .178]	Not Supported
H1b	EE → BI	-.008	.875	[-.108, .092]	Not Supported
H1c	SI → BI	.136	.024	[.018, .254]	Supported*
H1d	FC → BI	.059	.338	[-.062, .180]	Not Supported
H1e	HM → BI	.217	.014	[.044, .390]	Supported*
H1f	PV → BI	.505	<.001	[.352, .658]	Confirmed – Strongest
H1g	HB → BI	.023	.631	[-.071, .117]	Not Supported
H2	TR → BI	.106	.064	[-.006, .218]	Marginal

*z-test significant; bootstrap CIs include zero – interpreted as suggestive rather than conclusive. Bootstrap validation (1,000 resamples): only Price Value's path was confirmed as fully robust.

The Price Value Dominance Finding

A Striking Reversal

Blut et al. (2022) meta-analysis across 417+ UTAUT studies:
Performance Expectancy dominant (weighted $\beta = .39$)

Present study:

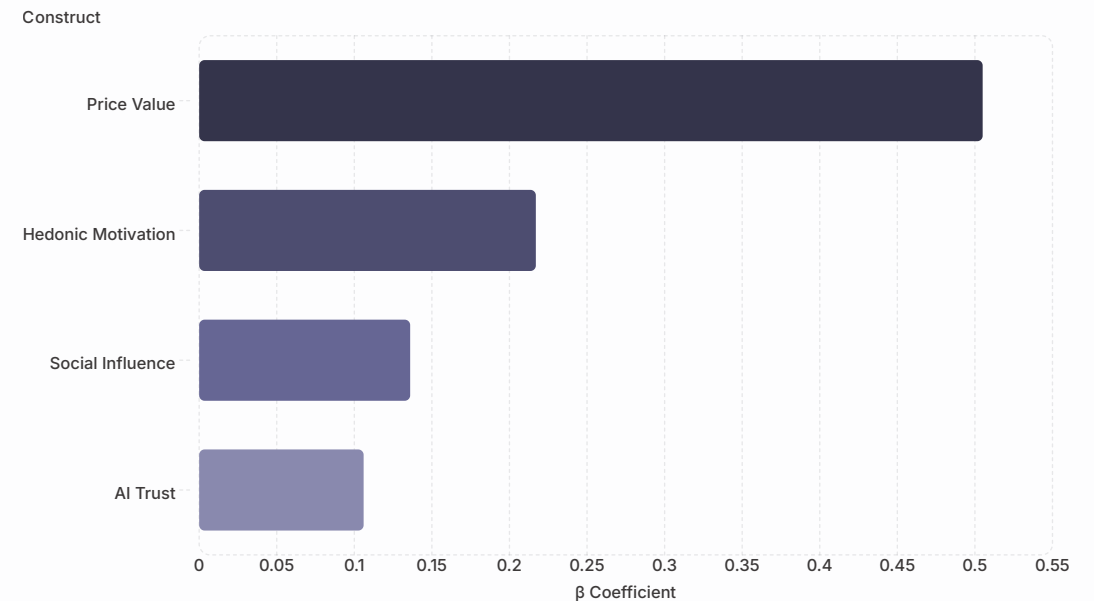
PE non-significant ($\beta = -.028$)

Price Value dominant ($\beta = .505$)

This reversal suggests AI tools represent a **distinct technology category** where cost-value outcomes have displaced perceived utility as the primary adoption driver.

Interpretive Logic

When users already regard AI's usefulness as a **baseline expectation** — supported by uniformly high PE means in this sample — adoption decisions shift to whether perceived benefits justify the financial investment.



AI Trust: Diagnostic Retention Rationale

1

Unique Diagnostic Capability

Trust provides organizational insight that other constructs cannot detect — identifying trust deficits that require targeted confidence-building interventions

2

Statistical Power Consideration

Detecting the marginal effect ($\beta = .106$) at 80% power requires $n > 600$; the present $N = 523$ yields ~68% power — the effect is theoretically relevant but underpowered

3

Parsimony vs. Utility

Model comparison favored the 7-factor model on parsimony ($\Delta AIC = +2.01$), but the 8-factor AIRS is retained as the recommended diagnostic instrument for organizational assessment

Moderation Effects

Experience Moderation (H3: Partially Supported)

Professional experience significantly moderated the **HM → BI path** ($\beta = .136$, $p = .007$). Professionals with 4+ years weight hedonic motivation more heavily as experience grows.

Usage-group comparison:

- Low-usage group: Performance Expectancy mattered ($\beta = .371$, significant)
- High-usage group: Price Value dominated ($\beta = .878$, significant)

Adoption drivers evolve with experience: new users focus on utility; experienced users focus on value.

Population Moderation (H4: Partially Supported)

Hedonic Motivation showed significant population moderation ($p = .041$):

- **Academic group:** HM $\beta = 0.449$ (strong positive)
- **Professional group:** HM $\beta = -0.301$ (negative)

Academics adopt AI partly for the enjoyment of exploration and learning. Professionals may view the "enjoyment" dimension as irrelevant or counterproductive to task-focused work.

Price Value remained dominant for **both groups**.

Behavioral Validation



Intention–Usage Correlation (H5: Supported)

Behavioral Intention was strongly associated with actual AI tool usage frequency: **Spearman $\rho = .69$, $p < .001$** , confirming AIRS measures a construct with real behavioral consequences.



Individual Tool Correlations

ChatGPT $\times$ BI: $\rho = .57$

Microsoft Copilot $\times$ BI: $\rho = .54$

Google Gemini $\times$ BI: $\rho = .52$



Role Differences (H6: Supported)

One-way ANOVA confirmed a clear hierarchy: **Leaders > Professionals > Academics** across tool breadth ($\eta^2 = .066$), usage frequency ($\eta^2 = .078$), and usage intensity ($\eta^2 = .058$).

Leaders demonstrated particularly elevated Microsoft Copilot usage (Cohen's $d = 1.14$ vs. Professionals), likely reflecting enterprise licensing and workflow integration.

User Typology: Three Segments

AI Enthusiasts — 31%

Early adopters with high engagement across all dimensions. Mean BI = 4.23. Deploy as **change champions** and peer influencers.

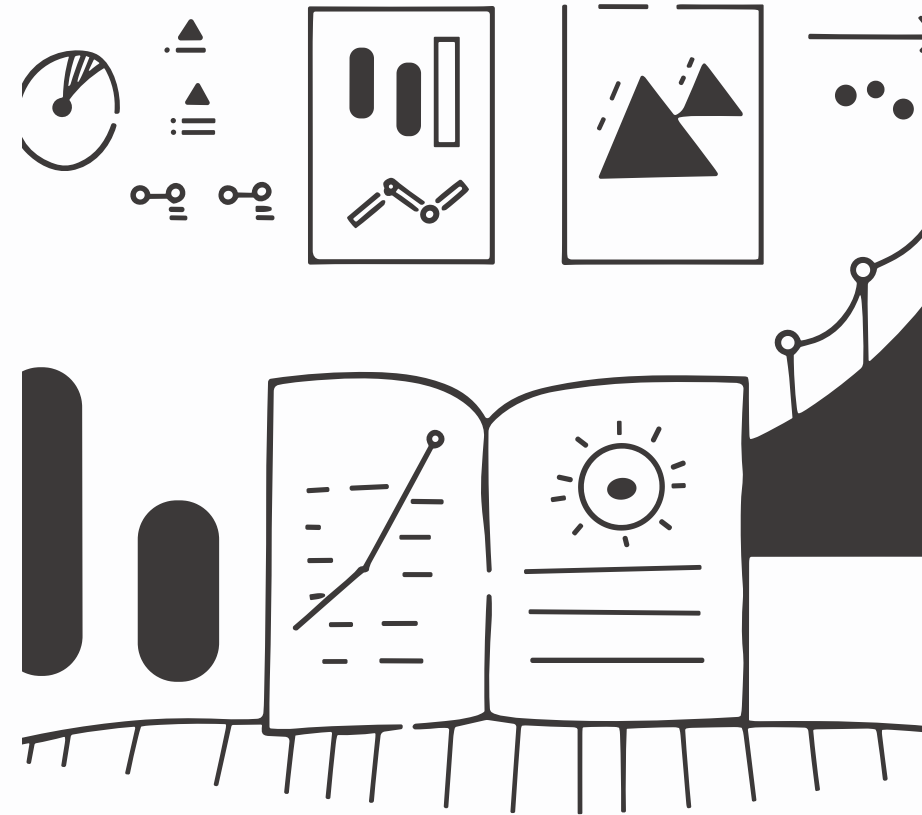
Moderate Users — 47%

Pragmatic users near population means. Respond to **value demonstrations** and social proof from peers.

AI Skeptics — 22%

Resistant users with below-mean readiness. Mean BI = 1.71. Require **trust-building and anxiety-reduction** before engagement strategies will succeed.

K-means cluster analysis ($k = 3$, silhouette = 0.271) explained **65.9% of BI variance** ($F(2,520) = 503.47$, $p < .001$, $\eta^2 = .659$), confirming a clear readiness gradient.



Hypothesis Outcome Summary

Category	Supported	Partial	Not Supported
UTAUT2 Core (H1a–g)	3 (43%)	0	4 (57%)
AI Extension (H2)	0	1 (marginal)	0
Moderation (H3–H4)	0	2	0
Behavioral (H5–H6)	2 (100%)	0	0
Total	5 (42%)	3 (25%)	4 (33%)

The Pattern Is the Finding

The distribution of supported vs. unsupported hypotheses reveals that **AI adoption follows a different predictor structure** than traditional technology adoption — with cost-value perceptions displacing utility perceptions as the primary driver.



Chapter 5

Theoretical Contributions

Contribution 1 & 2: UTAUT2 Extension & PV Dominance

First Empirical Validation of UTAUT2 for AI

This is the first study to empirically validate UTAUT2 for AI tool adoption with a purpose-built psychometric instrument. The 8-factor structure demonstrates UTAUT2's architecture applies to AI contexts, but predictor importance shifts substantially — four traditional predictors (PE, EE, FC, HB) were non-significant.

AI as a Distinct Technology Category

Price Value dominance ($\beta = .505$) challenges decades of UTAUT research positioning Performance Expectancy as the primary driver. In AI contexts, utility may function as a **baseline expectation** rather than a differentiating factor, with adoption shaped more strongly by perceived cost-benefit ratios.

Contributions 3–5: Career Stage, Typology & Industry Validation

1 Career Development Integration

The HM → BI moderation ($p = .007$) and population moderation (Academic $\beta = 0.449$ vs. Professional $\beta = -0.301$) establish that adoption mechanisms are **career-stage dependent** — a novel bridge between career development theory and technology acceptance.

2 Three-Segment User Typology

The empirically derived typology (Enthusiasts, Moderate Users, Skeptics) explained **65.9% of BI variance**, providing a segmentation framework grounded in validated psychometric dimensions rather than ad hoc categorization.

3 Converging 2025 Industry Validation

McKinsey (N = 1,993), BCG, MIT Media Lab, Gartner, Deloitte, and Capgemini each reported patterns broadly consistent with AIRS findings — including centrality of perceived value, heterogeneous readiness, and limits of one-size-fits-all strategies.

Contribution 6: Four Adoption-Value Gap Mechanisms



Value Communication Misalignment

Organizations emphasize capability rather than ROI, failing to address the dominant PV predictor



Heterogeneous Readiness

Treating the workforce as monolithic when 22% are Skeptics requiring fundamentally different intervention



Neglected Affective Barriers

Anxiety and hedonic motivation are ignored in favor of rational-utitarian framings



Context-Inappropriate Frameworks

Generic technology adoption strategies fail to account for AI's distinctive characteristics as a technology category

Chapter 6

Practical Implications

The AIRS findings translate directly into evidence-based recommendations for organizations deploying AI tools, AI tool designers, trainers, and policymakers.



For Organizations: Eight Practitioner Recommendations

01

Diagnose Before Prescribing

Conduct baseline AIRS assessment before designing AI adoption initiatives

03

Attend to Affective Barriers

Hedonic motivation and anxiety influence adoption beyond rational calculations

05

Differentiate by Experience

Novice and experienced users respond to different adoption drivers

07

Design for Accessibility

Universal design principles reduce anxiety-driven non-adoption among users with disabilities

02

Lead with Value

Frame AI in terms of cost-benefit returns (PV dominance), not capability showcases

04

Leverage Social Influence

Peer champions and leadership endorsement are the third significant predictor (SI $\beta = .136$)

06

Monitor Trust Continuously

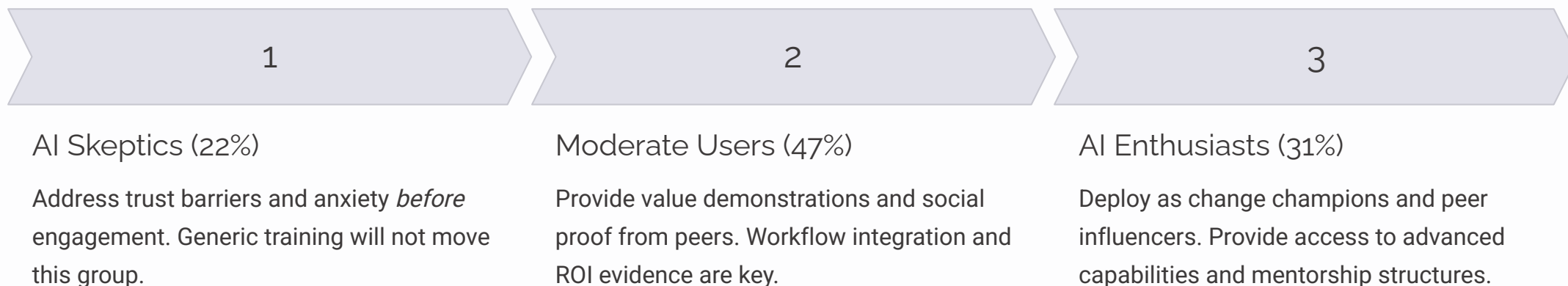
Trust functions as a diagnostic indicator and necessary condition, not just a predictor

08

Segment the Workforce

Any single intervention strategy will fail for at least one-third of the workforce

Segment-Specific Intervention Strategy



For AI tool designers: manage pricing and perceived value explicitly, and treat accessibility as an adoption condition — not merely an inclusion issue.

Special Section

Neurodiversity & Disability Insights

The AIRS dataset includes **68 participants (13.0%)** who self-reported a disability — sufficient for exploratory between-group analysis and among the most practically salient findings in the dissertation.



Disability–Anxiety Association: Core Finding

Statistically Significant Effect

Participants with disabilities reported significantly higher AI-related anxiety than non-disabled participants:

- Disability group: $M = 2.89$, $SD = 1.08$
- Non-disability group: $M = 2.48$, $SD = 0.96$
- $t(521) = 2.76$, $p = .006$, Cohen's $d = 0.36$

Notably, this finding emerged from AI Anxiety items that were **subsequently excluded** from the final AIRS-16 due to inadequate psychometric properties — suggesting the true effect may be larger than observed.

Coherent Disadvantage Profile

Across all 8 UTAUT2 constructs, participants with disabilities showed **consistently lower scores** on every positive adoption construct and higher anxiety — a uniform directional pattern constituting coherent disadvantage rather than random noise.

Hedonic Motivation also reached significance ($d = .28$, $p = .030$): current AI tools are **less enjoyable** for users with disabilities — pointing to accessibility failures in existing AI products.

Disability Findings: Four Interpretive Threads

Anxiety Reflects Lived Experience

For users who have repeatedly encountered inaccessible interfaces or algorithmic bias, elevated anxiety about AI is a **rational response to historical exclusion** — not a deficit to be corrected.

AI Anxiety Construct Needs Redesign

The AIRS-28 expansion should prioritize redesigning anxiety items using neurodiversity-informed language distinguishing displacement anxiety from accessibility-barrier anxiety.

HM Suppression Is a Design Problem

The significant HM gap ($d = .28$) implies current AI tools are less enjoyable for users with disabilities — pointing to accessibility failures, not user deficits.

Universal Design as Adoption Accelerator

Making AI tools more accessible reduces anxiety-driven non-adoption broadly — the **"curb cut effect"** applied to AI adoption benefits the entire user population.

Chapter 7

Limitations

Methodological

- **Cross-sectional design:** Prevents causal inference about direction of relationships or how adoption intentions evolve over time
- **Self-reported intention:** BI-Usage correlation ($\rho = .69$) confirms meaningful association, but intention does not perfectly predict behavior
- **Panel sampling:** Centiment panel may not perfectly represent the broader U.S. population

Measurement & Scope

- **Two items per factor:** Constrains reliability and contributes to elevated inter-factor correlations
- **Four dropped constructs:** Voluntariness, Explainability, Ethical Risk, and AI Anxiety warrant item redesign and re-investigation
- **Western sample:** All U.S.-based participants; AI adoption patterns may differ substantially in non-Western contexts

Chapter 8

Future Research Roadmap



Four-Phase Research Agenda



Phase 1: AIRS-28 Development (2026–2027)

Expand from 16 to 28 items (3–4 per factor) to improve reliability and resolve discriminant validity concerns. Redesign Explainability, Ethical Risk, and AI Anxiety items. Target: eliminate all HTMT violations.



Phase 3: Intervention Framework Development (2028–2029)

Design and test AIRS-informed organizational change programs targeted at each user segment. Test whether AIRS-guided interventions outperform generic AI training approaches.



Phase 2: Longitudinal & Cross-Cultural Validation (2026–2028)

12-month longitudinal panel to track readiness profile evolution. Simultaneously validate AIRS across non-Western cultural contexts to establish cross-cultural measurement invariance.



Phase 4: Comprehensive Prediction System (2029+)

Integrate AIRS with behavioral data, organizational context measures, and industry-specific moderators to build a comprehensive AI readiness prediction and monitoring system.

Seven Scholar Recommendations

- **Larger samples** ($n > 600$) to increase power for multi-group and moderation analyses
- **Improved AI-specific measures** with 3–4 items per dimension to resolve HTMT violations inherent in 2-item scales
- **PV dominance investigation** – replicate and explain why Price Value overtakes Performance Expectancy specifically in AI contexts
- **Career development integration** – study how adoption mechanisms change across career stages (student → early → mid → senior)
- **Typology validation** – confirm the three-segment structure in independent samples and test whether typology membership predicts actual adoption behavior
- **Cross-cultural replication** – establish measurement invariance across non-Western contexts
- **Longitudinal designs** – move beyond cross-sectional snapshots to track readiness trajectories and determine whether AIRS scores predict future behavior

Appropriate Reliance & 2026 Contextual Developments

Calibrated AI Reliance

A distinctive future direction concerns the paradox that optimizing AI adoption may inadvertently increase over-reliance. The AETHER group's **Calibrated AI Reliance (CAIR)** and Calibrated Self-Reliance (CSR) constructs suggest the goal should be *appropriate* adoption — knowing when to use AI and when to rely on human judgment.

This reframing from "more adoption is better" to "calibrated adoption is optimal" represents an important evolution of the research program, also addressed in *The Life of Alex Finch* (AlexBooks, March 2026).

Three 2026 Landscape Shifts

- **DeepSeek R1:** Open-source frontier capabilities change the PV calculation entirely — no longer exclusive to well-resourced organizations
- **Agentic AI:** Systems operating with greater autonomy raise new trust and control questions beyond the original AIRS scope
- **EU AI Act:** Enforcement timeline introduces regulatory compliance as a new adoption consideration absent from the current UTAUT2 framework



Chapter 9

AIRS Enterprise Platform

AIRS Enterprise (airs.correax.com) translates the validated AIRS-16 instrument into a production SaaS platform – delivering practical assessment and remediation tools to organizations while enabling longitudinal data collection at scale.

Platform Core Capabilities



5-Minute Assessment

Interactive AIRS-16 with progress tracking, save/resume, and smart follow-up questions. Composite AIRS score (8–40) with demonstrated validity ($r = .876$).



AI-Personalized Guides

Streaming LLM-generated action plans (3–5 pages) tailored to individual profile and career context via Azure OpenAI GPT-4o-mini. Available in 29 languages.



Results Visualization

Animated gauge, 8-axis construct radar chart, and typology badge with actionable context. Three-segment classification at 94.5% accuracy.



Organization Management

Team analytics, department/role segmentation, CSV exports, impact reporting, and longitudinal pre/post assessment comparison with trend analysis.

Technology Stack: Next.js 16.1, React 19, Prisma ORM with PostgreSQL, Microsoft Entra ID authentication, Azure App Service Premium (P2v3) with VNet integration and managed identity — no stored secrets.

Three Intervention Frameworks by Typology

TRANSFORM Framework

Skeptics (AIRS ≤ 20)

Trust-building through graduated autonomy — starting with low-risk AI tasks, demonstrating reliability through small wins, progressively expanding scope. Emphasizes explainability and human-in-the-loop safeguards.

REDESIGN Framework

Moderate Users (AIRS 21–30)

Workflow integration and ROI demonstration — identifying specific processes where AI adds measurable value, providing social proof through peer success stories, building competence through structured skill development.

AMPLIFY Framework

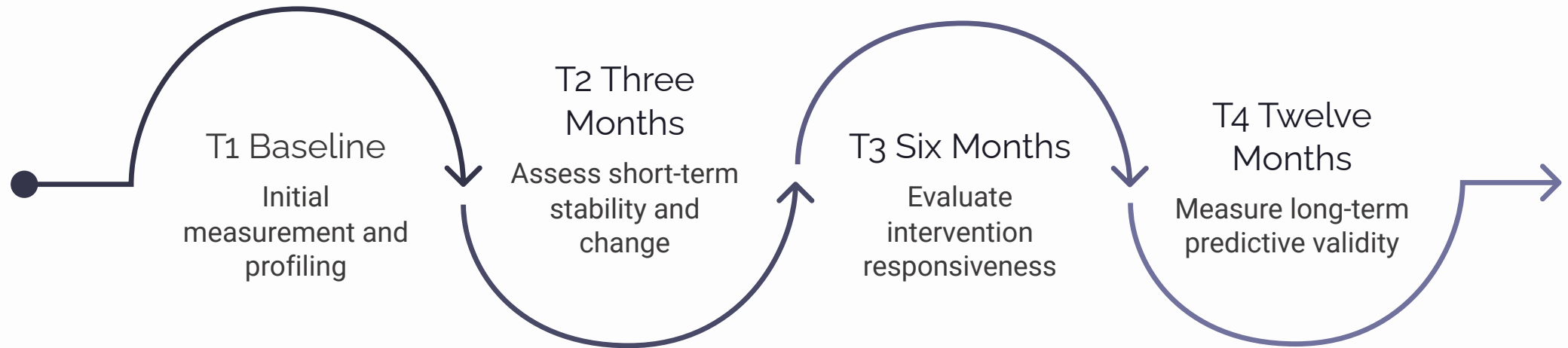
Enthusiasts (AIRS > 30)

Champion network activation — leveraging high-readiness individuals as change agents, providing access to advanced AI capabilities, and creating mentorship structures that accelerate adoption across the organization.

Construct-Specific Interventions

Construct	Barrier Indicator	Evidence-Based Intervention
Price Value	Low perceived ROI	TCO analysis workshops, pilot-to-proof ROI tracking, phased investment models
Hedonic Motivation	Low engagement	Gamification elements, AI sandboxes for exploration, social learning communities
Social Influence	Insufficient advocacy	Peer champion programs, leadership endorsement campaigns, success story sharing
AI Trust	Reliability concerns	Explainability tools, human-in-the-loop with graduated autonomy, transparent error reporting
Effort Expectancy	Usability barriers	Guided onboarding, contextual help systems, progressive complexity
Habit	Low integration	Default AI-assisted workflows, reminder systems, routine integration coaching

Longitudinal Research Design: AIRS Enterprise



Every organizational deployment generates structured data (8 construct scores, composite AIRS score, typology classification, demographic context) feeding directly into a normative benchmarking database targeting N = 300–500+ per population group across 8+ target groups (2026–2028).

Research-Practice Bridge

Closing the Implementation Gap

Instead of the familiar pattern in which an instrument is developed, published, and then rarely operationalized, AIRS pairs the validated instrument with a platform that **evolves alongside it**. Every organizational deployment strengthens the normative database; every research finding improves intervention recommendations.

The medium-term strategy (1–3 years) focuses on building a benchmark database. The longer-term goal (3–5 years) is to establish AIRS as a recognized AI readiness standard through academic use, organizational partnerships, and integration with enterprise HR and LMS systems.

Three Complementary Paths

- **AIRS-16 Instrument:** Validated psychometric tool for academic and organizational use
- **AIRS Enterprise Platform:** Production SaaS delivering assessment, personalization, and longitudinal tracking
- ***The Life of Alex Finch*** (AlexBooks, March 2026): Public-facing narrative on appropriate reliance, over-reliance, and preservation of human judgment

Together, these create complementary paths for advancing the research across academic, organizational, and public audiences.